



From lecture session 8: We explored the importance of marine sediments. Now, we will move from marine geology and look at the chemistry of the ocean.

Lecture Session 9

1. Water's atoms and molecules and implications to its physical properties;
2. Water's thermal capabilities;

Icebreaker

Some of the reasons why coastal towns are so pleasant are the constant breeze and the comfortable temperatures all year long.

- Can you explain how the ocean and these amazing conditions are related?



San Diego, CA

Icebreaker

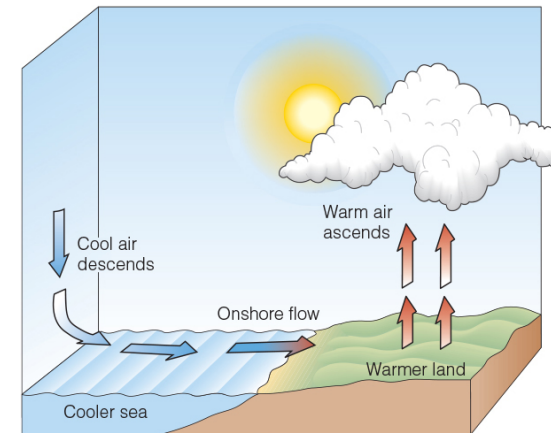
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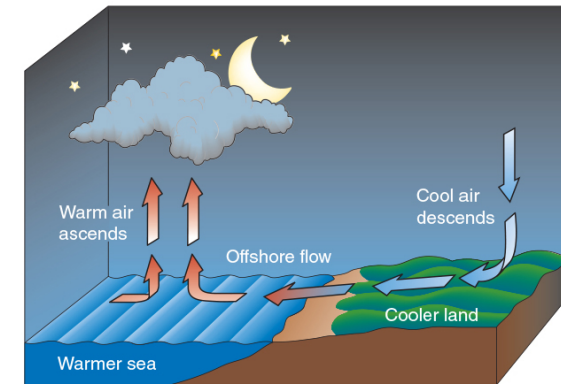


San Diego, CA

The answer is related to water properties: Ex. High Heat Capacity



a In the afternoon, the land is warmer than the ocean surface, and warm air rising from the land is replaced by an onshore sea breeze.



b At night, as the land cools, the air over the ocean is now warmer than the air over the land. The ocean air rises. Air flows offshore to replace it, generating an offshore flow (a land breeze).

Figure 8.20 The flow of air in coastal regions during stable weather conditions.

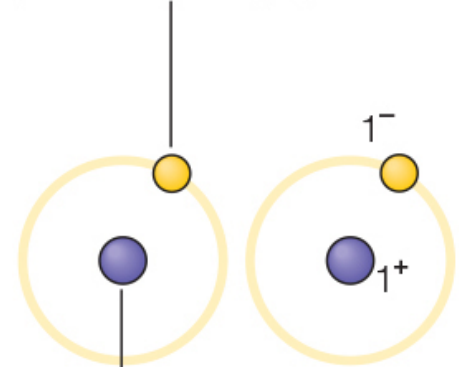
The Water Molecule: H₂O

Protons: + charge
Electrons: - charge

Covalent Bonds = connection
due to sharing of electrons

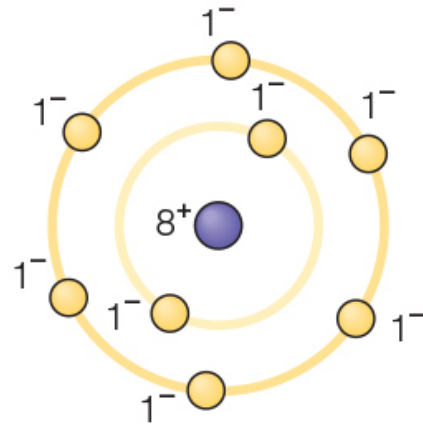
Two hydrogen atoms . . .

Electron
(1⁻ unit of charge)

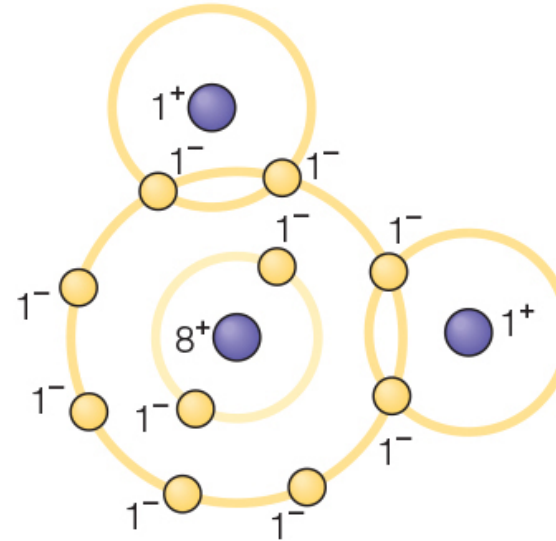


Nucleus
(1⁺ unit of charge)

share their electrons
with one oxygen atom . . .



to form a water molecule held
together by covalent bonds . . .



which acts as if it has
negative and positive ends.

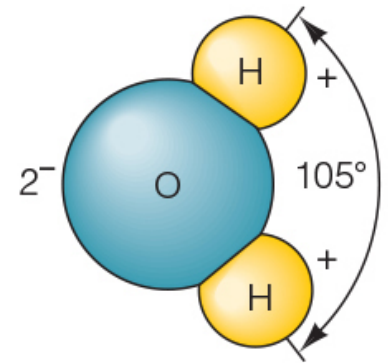


Figure 6.3 The formation of a water molecule. Atoms are most stable when they have a full outer shell of electrons. Hydrogen is most stable when it has two electrons in its outer shell and the larger oxygen atom (like many of the atoms relevant to our discussion of oceanography), is typically most stable with a total of 8 electrons in its outer shell. Since oxygen only has six electrons in its outer shell (because the first two of its eight electrons must fill the inner shell), oxygen bonds to two different hydrogen atoms by sharing hydrogen's electrons so that each atom has a full outer shell.

The Water Molecule: H₂O

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H₂O molecule configuration leaves the H's nucleus (protons) exposed. In contrast, the O's electrons on the other side are also exposed forming a Polar molecule. Therefore, like a magnet, it has a positive and negative charge

Two hydrogen atoms . . .

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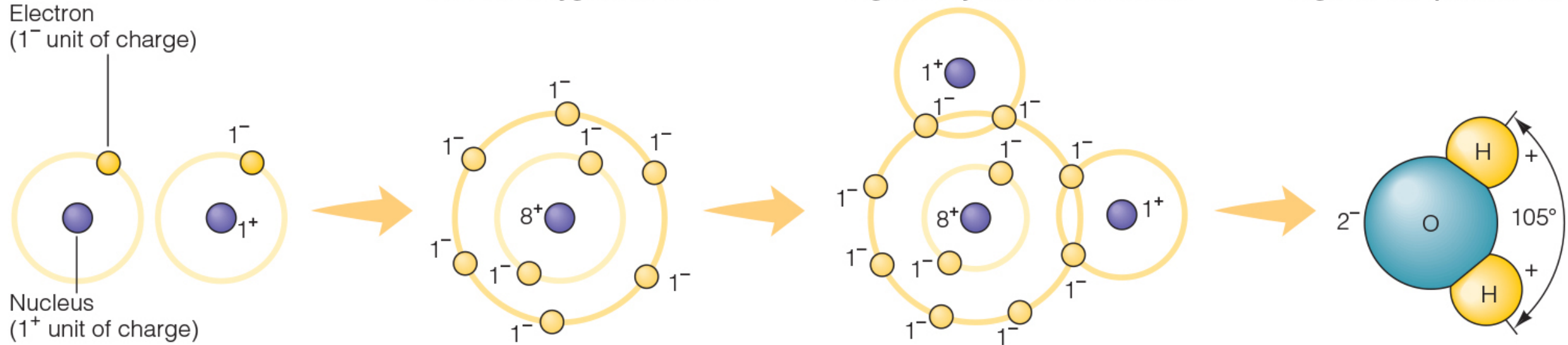
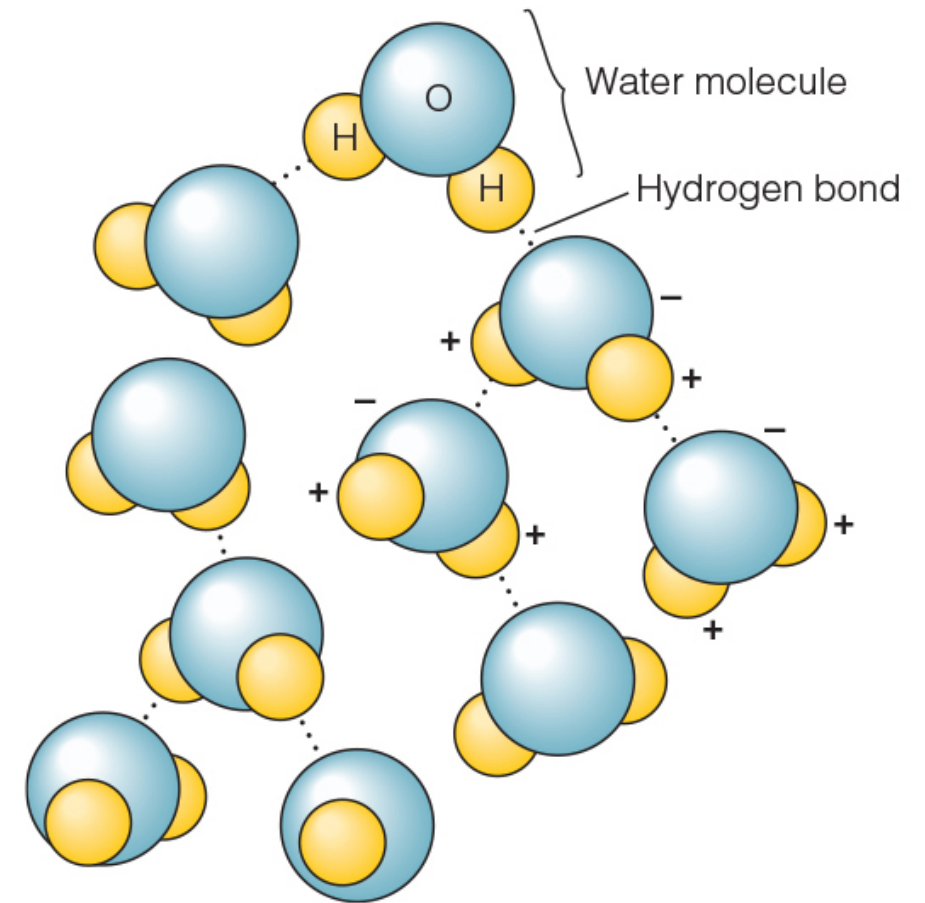


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Due to its polarity water molecules are capable of forming Hydrogen Bonds

Hydrogen Bonds: They link water molecules together by electrostatic forces

Hydrogen Bonds are about 5-10% as strong as covalent bonds within the molecule, forming a webwork of molecules with very peculiar chemical and physical properties



- a** Hydrogen bonds in liquid water. The attractions between adjacent polar water molecules form a webwork of hydrogen bonds. These bonds are responsible for cohesion and adhesion, the properties of water that cause surface tension and wetting. Hydrogen bonds among water molecules also make it difficult for individual molecules to escape from the surface.

The polarity also allows the water to be a powerful solvent

The power of Hydrogen Bonds

- Cohesion – water molecules stick to each other, creating surface tension
- Adhesion – water molecules stick to other substances and make them wet

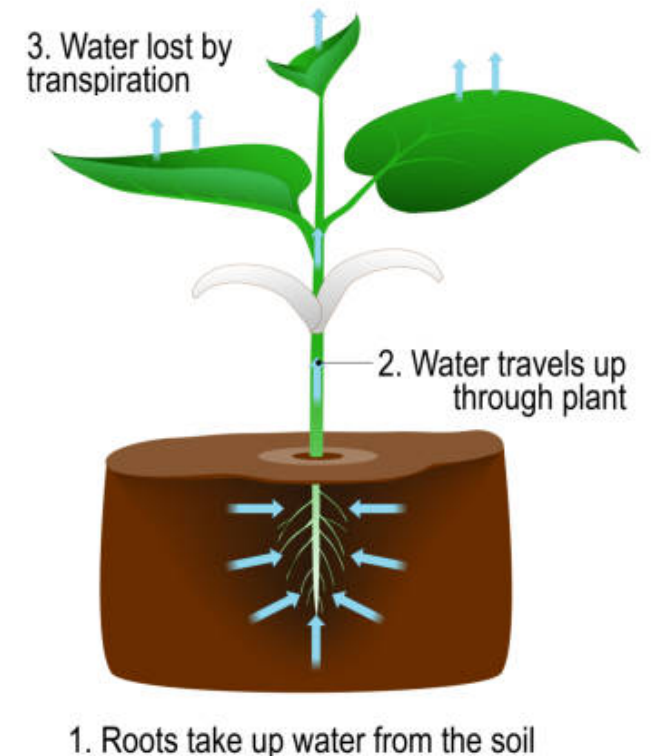
Causes of capillary action: From absorbent paper towels to plants hydration.



Prokuronov Andrey/Shutterstock.com

b Surface tension allows small bugs to be able to walk on water.

Figure 6.4 Hydrogen bonds and surface tension.



The power of Hydrogen Bonds

- Cohesion – water molecules stick to each other, creating surface tension
- Adhesion – water molecules stick to other substances and make them wet
- Molecules vibration push the neighboring ones absorbing more red light, allowing more blue light to get to us.

Iceberg



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- **Water's state: Ice (solid), liquid water, water vapor (gas). Without the hydrogen bounds, liquid water would never be possible (gas or solid).**

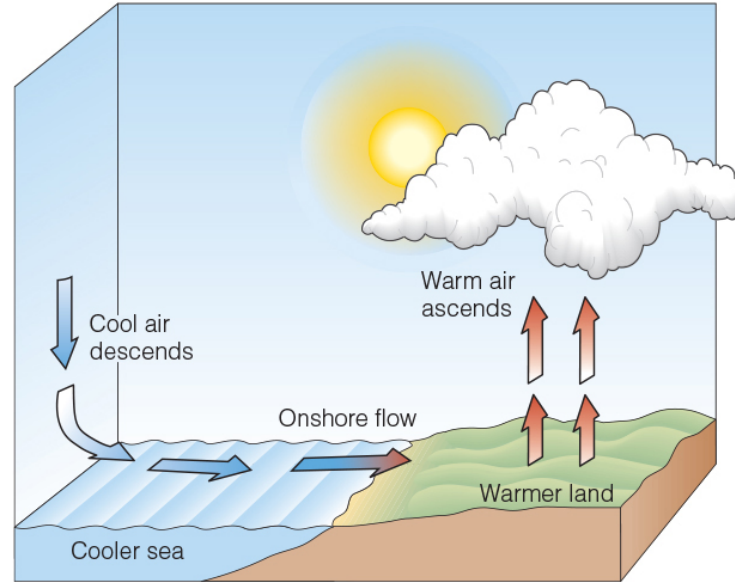
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- Water's state: Ice (solid), liquid water, water vapor (gas). Without the hydrogen bounds, liquid water would never be possible (gas or solid).
- **High Heat Capacity = It requires a lot of energy (heat) to change the water's temperature.**

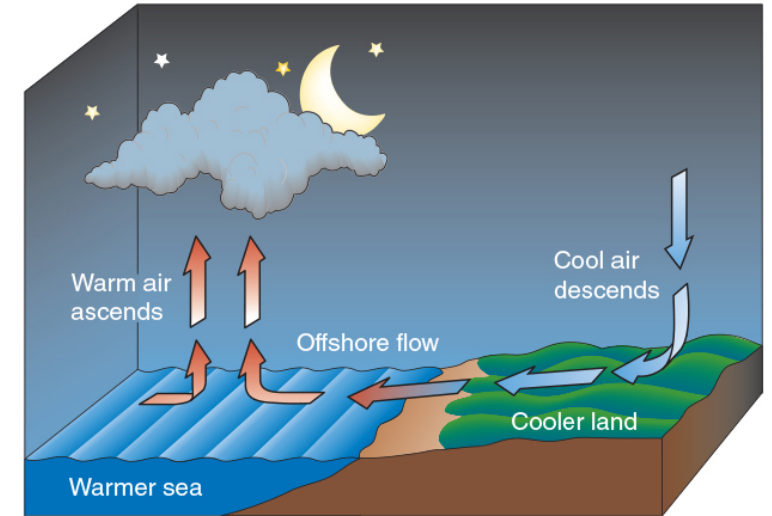
This property is very important for the ocean circulation, climate, and life on Earth

Thinking moment

What is the connection between heat capacity and the sea/land breeze?



- a** In the afternoon, the land is warmer than the ocean surface, and warm air rising from the land is replaced by an onshore sea breeze.



- b** At night, as the land cools, the air over the ocean is now warmer than the air over the land. The ocean air rises. Air flows offshore to replace it, generating an offshore flow (a land breeze).

Figure 8.20 The flow of air in coastal regions during stable weather conditions.

☐ Breezes are possible because the ocean's and land's heat capacities are similar

☐ The high heat capacity of the water is responsible for the sea breeze only

☐ The difference of the ocean's and land's heat capacities force the breezes

☐ The low heat capacity of the land is responsible to the land breeze only

An aerial photograph of a tropical coastline. The top half of the image shows deep blue ocean water. Below this is a narrow strip of white sand beach, followed by a shallow lagoon with turquoise water and visible coral reefs. The bottom half of the image shows a vast expanse of green, vegetated land. The text "Thermal Properties of Water" is overlaid in white, centered horizontally across the middle of the image.

Thermal Properties of Water

Before we dive into the properties: What is the difference between Heat and Temperature?

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- **Heat – energy produced by vibration of atoms/molecules**
- **Temperature – object's response to input or removal of heat (or vibration). Measure of how much energy there is in the system.**

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- **Heat – energy produced by vibration of atoms/molecules**
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- Heat capacity = Amount of energy necessary to change the temperature of a certain mass of a material.
- **Water has high heat capacity (same as previous slide in other words) = Can absorb or release large amounts of heat with no small change in temperature**

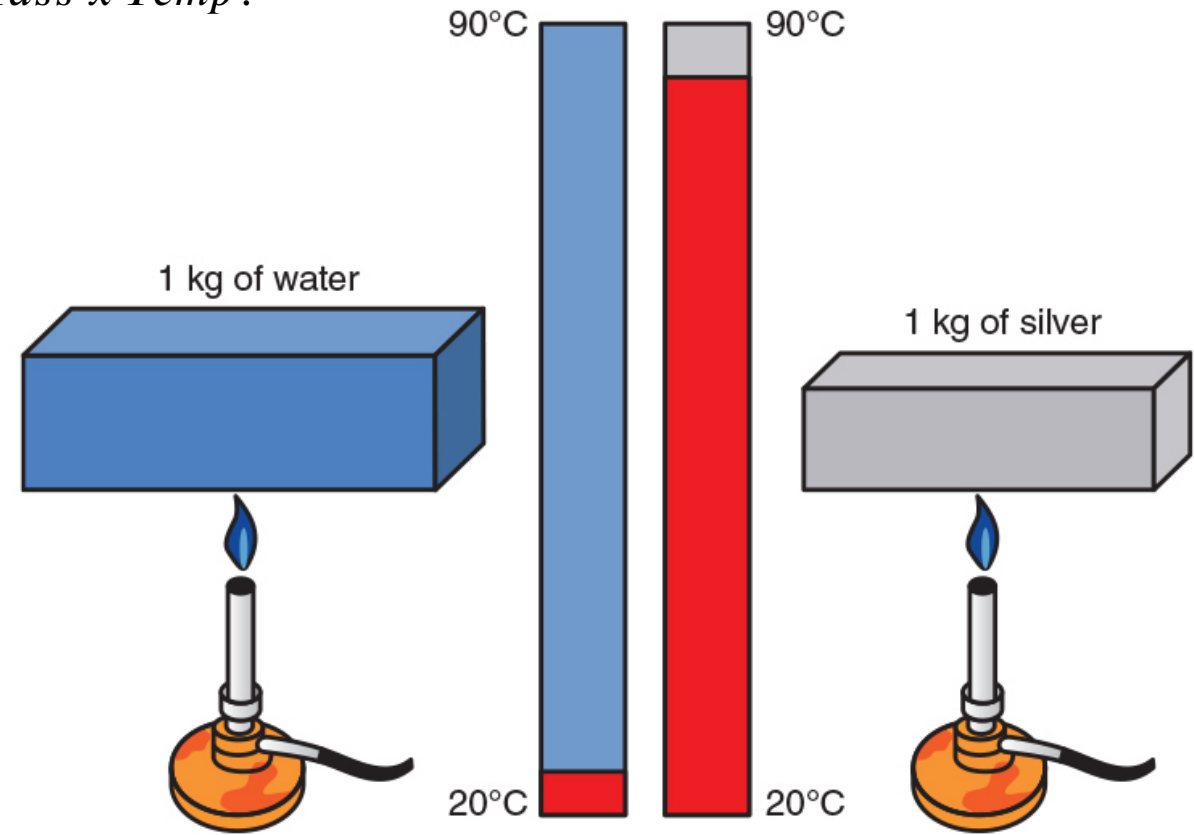
Water Especial Thermal Characteristics

Substance	Specific Heat Capacity in calories/gram/°C
Silver	0.06
Granite	0.20
Aluminum	0.22
Alcohol (ethyl)	0.30
Gasoline	0.50
Acetone	0.51
Pure water	1.00
Ammonia (liquid)	1.13

- a** Heat capacity is a measure of the heat required to raise the temperature of a substance by 1°C (1.8°F). When comparing the heat capacity of different substances per unit mass, it is often referred to as the *specific heat capacity*, or *specific heat*, of a substance. Notice how little heat is required to raise the temperature of 1 gram of silver 1° in comparison with 1 gram of water.

Figure 6.5 Heat capacity of common substances.

$$\frac{\text{Energy}}{\text{Mass} \times \text{Temp.}}$$



- b** Temperature change in one kilogram of water and silver when exposed to the same amount of heat.

Quick Concept check:

1. Which material at 20° C has more internal energy (i.e., molecules vibration)?
- 1 kg of Silver
1 kg of Water

2. Using the concept of Heat Capacity (C_p), can you confirm your answer? Calculate the energy of both materials.

$$\text{Silver } C_p = 0.06 \frac{\text{calories}}{\text{gram} \cdot ^\circ\text{C}}$$

$$\text{Pure Water } C_p = 1.0 \frac{\text{calories}}{\text{gram} \cdot ^\circ\text{C}}$$

3. Are there any connections between the physical state of each material at 20° C (68° F) and the amount of vibration of the molecules?

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$$\text{Silver } C_p = 0.06 \frac{\text{calories}}{\text{gram} \cdot ^\circ\text{C}}$$

$$\text{Silver energy} = 1200 \text{ calories}$$

$$\text{Pure Water } C_p = 1.0 \frac{\text{calories}}{\text{gram} \cdot ^\circ\text{C}}$$

$$\text{Water energy} = 20000 \text{ calories}$$

3. Are there any connections between the physical state (solid, liquid, gas) of each material at 20° C (68° F) and the amount of vibration of the molecules?

At 20° C water is liquid and silver is solid. The rigid structure of solid materials is given by the locking of molecules into a strict geometrical order (almost no motion). In liquids and gases, the molecules are “free” to move around.

Latent vs Sensible Heat

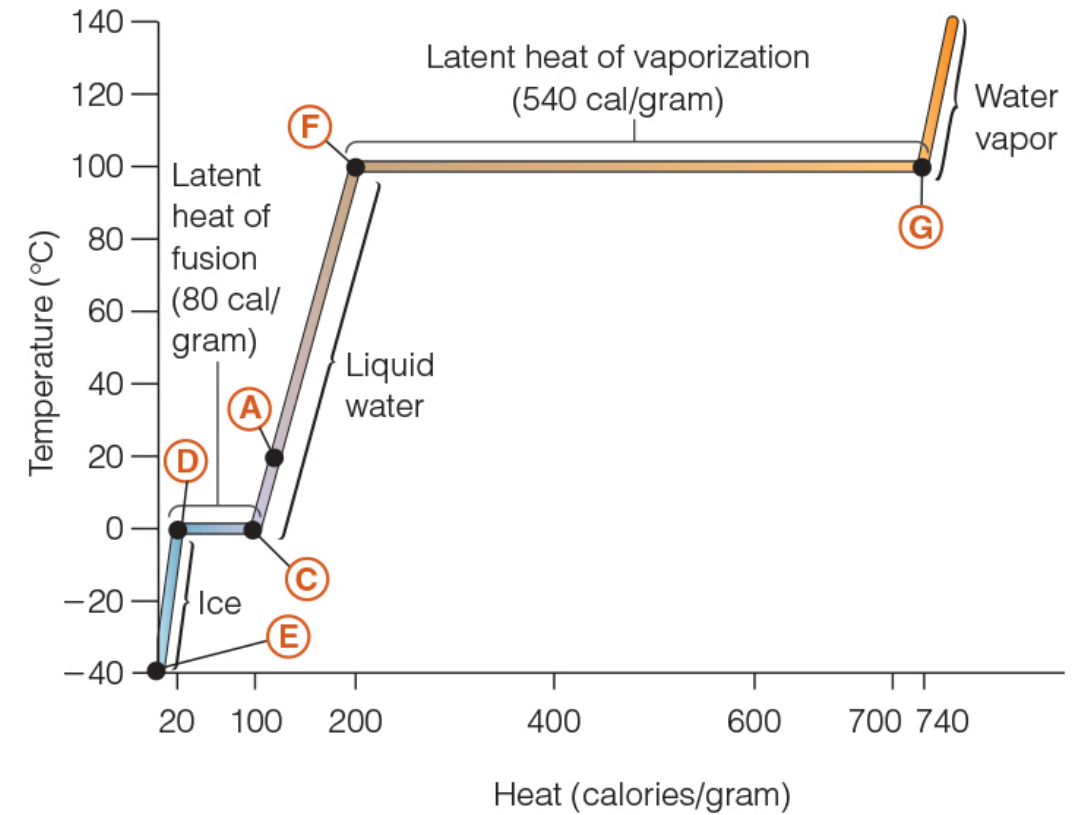


Figure 6.9 A graph of temperature versus heat as water freezes, melts, and vaporizes. The horizontal line between points **C** and **D** represents the latent heat of fusion, when heat is being added or removed but the temperature is not changing. The horizontal line between points **G** and **F** represents the latent heat of vaporization, when heat is being added or removed but the temperature is not changing. (Note that points **A–E** on this graph are the same as those in Figure 6.6.)

Latent vs Sensible Heat

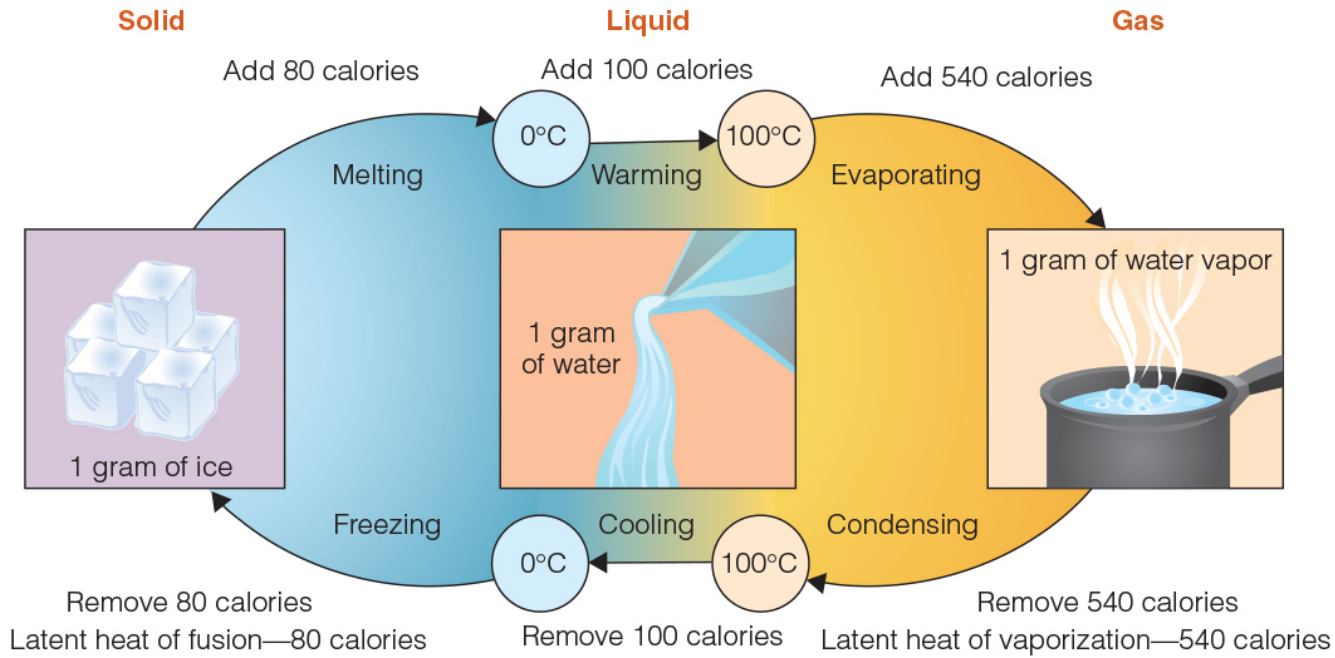


Figure 6.11 We must add 80 calories of heat energy to change a gram of ice to liquid water. After the ice is melted, about 1 calorie of heat is needed to raise each gram of water by 1°C. But 540 calories must be added to each gram of water to vaporize it—to boil it away. The process is reversed for condensation and freezing.

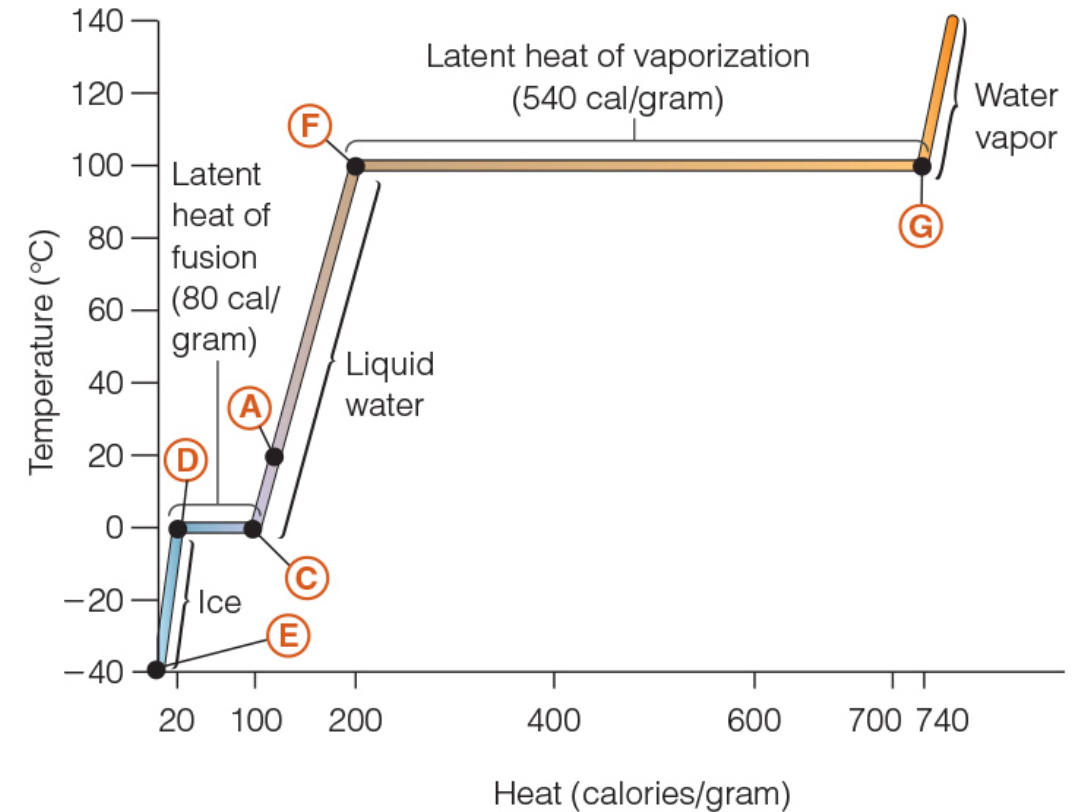
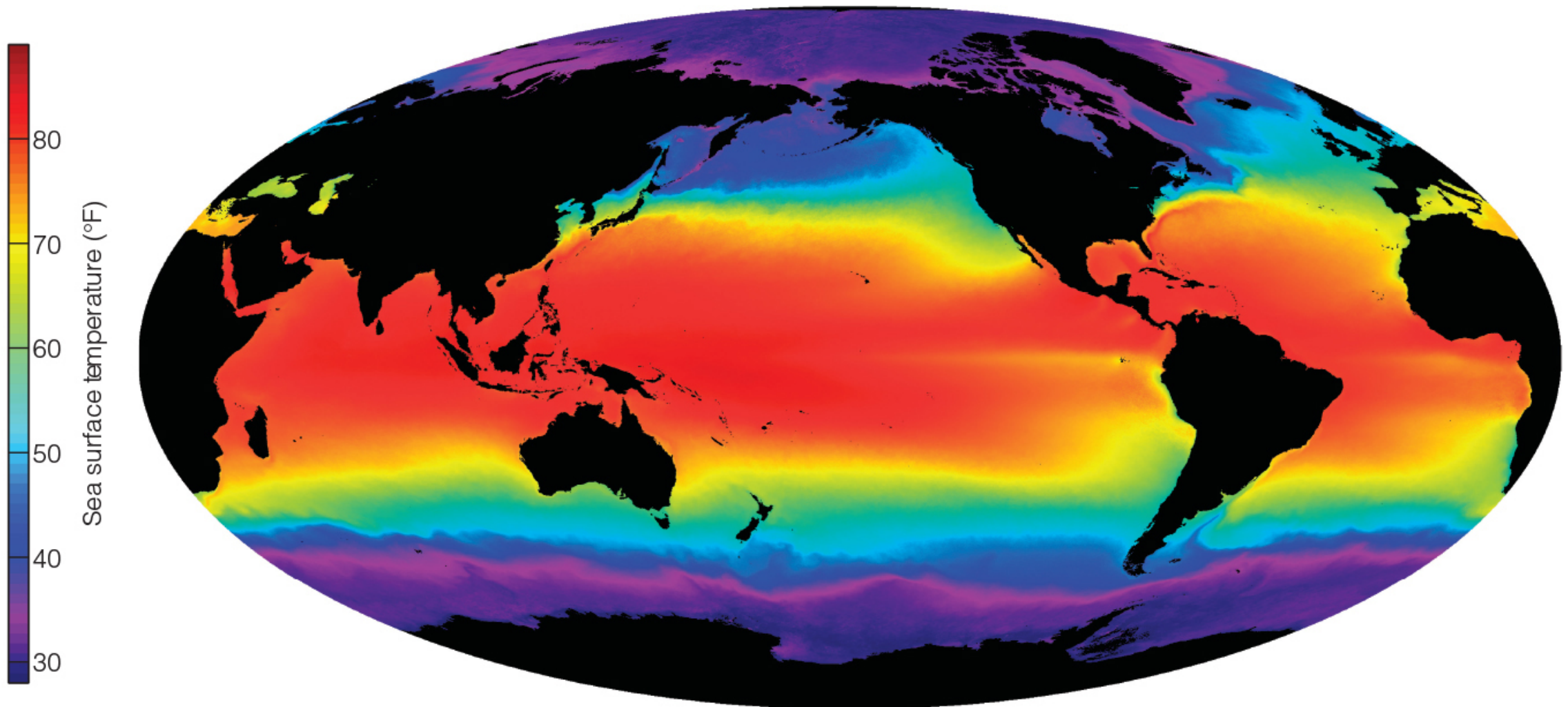


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Why do we care about such properties?



Typical Sea Surface Temperatures observed using satellites

Ocean Moderates Global Climate (even over the continents, remember the icebreaker exercise!)

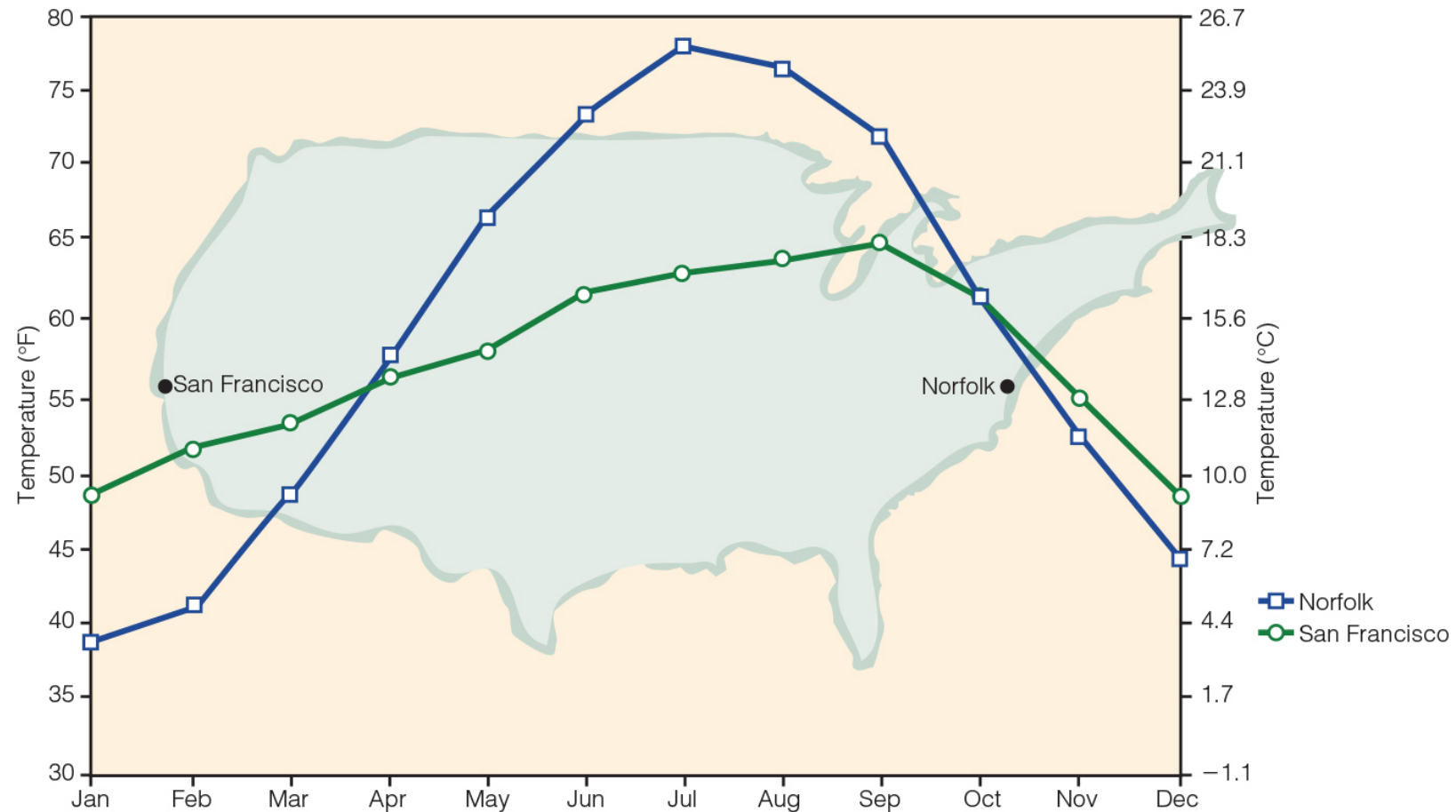


Figure 6.13 San Francisco, California, and Norfolk, Virginia, are on the same line of latitude, yet San Francisco is warmer in the winter and cooler in the summer than Norfolk. Part of the reason is that wind tends to flow from west to east at this latitude. Thus, air in San Francisco has moved over the ocean whereas air in Norfolk has approached over land. Water does not warm as much as land in the summer, nor cool as much in winter—a demonstration of thermal inertia.

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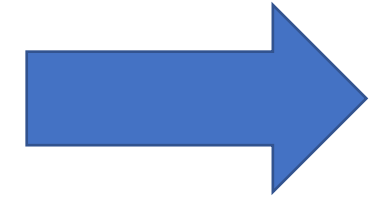
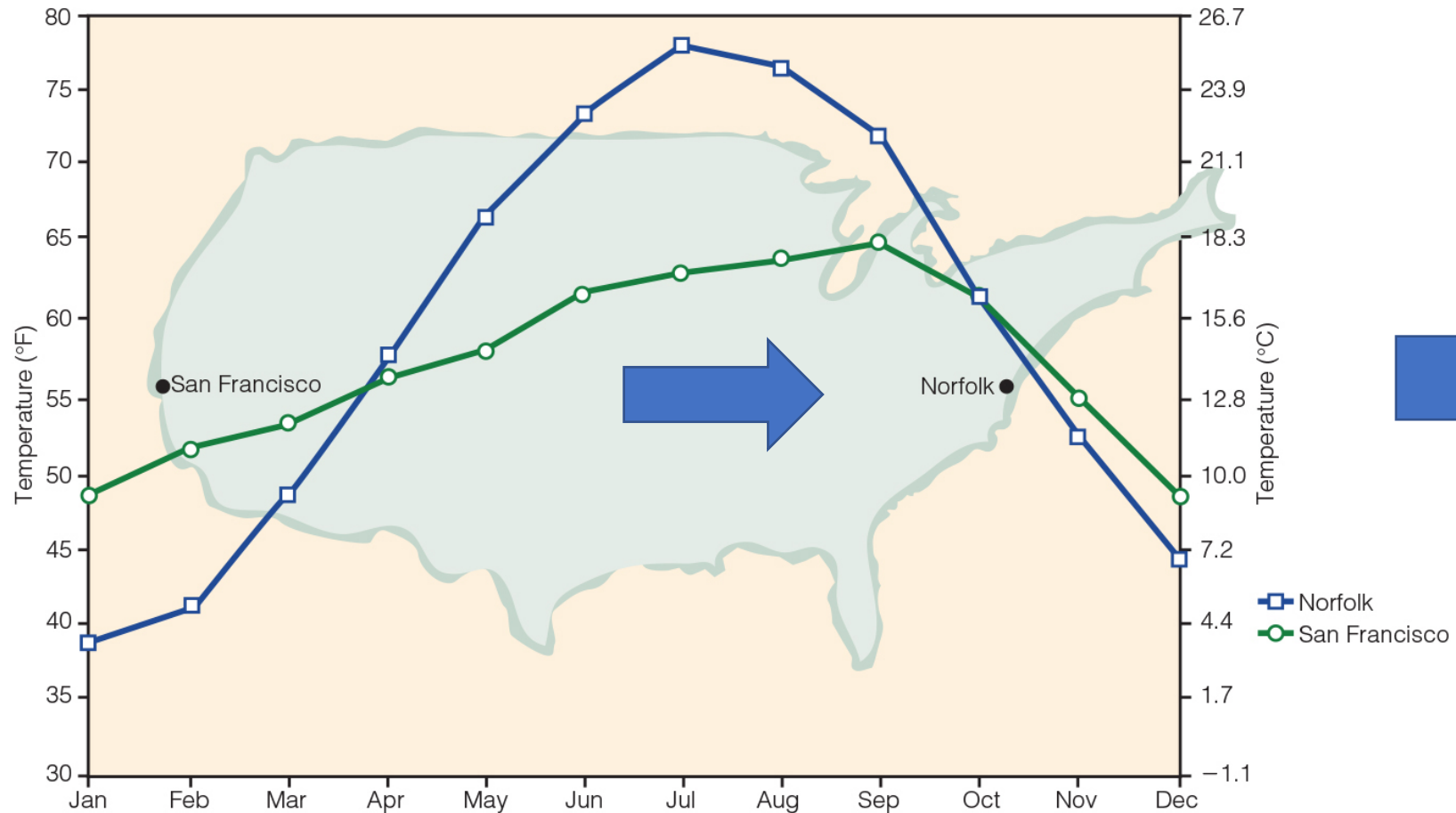


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